

第四单元 化学实验与应用 (Unit 4 Chemistry Experiments and Applications)

第二章 常见气体制备与检验 (Chapter 2 Preparation and Identification of Common Gases)

1. 下列关于氮元素及氮气性质的说法中错误的是 (Which of the following statements about nitrogen element and nitrogen gas properties is incorrect?) ()

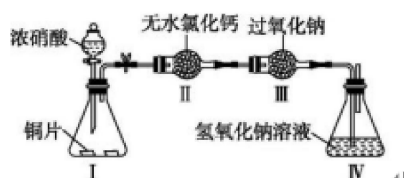
- A. 氮在自然界中既以游离态存在又以化合态存在 (Nitrogen exists in both free state and combined state in nature)
B. 通常情况下, 氮气性质很不活泼 (Under normal conditions, nitrogen gas is very inactive)
C. 在反应 $3\text{Mg} + \text{N}_2 \xrightarrow{\text{点燃}} \text{Mg}_3\text{N}_2$ 中 N_2 作氧化剂 (In the reaction $3\text{Mg} + \text{N}_2 \xrightarrow{\text{ignition}} \text{Mg}_3\text{N}_2$, N_2 acts as an oxidizing agent)
D. 放电条件下氮气可与氧气反应生成二氧化氮 (Nitrogen gas can react with oxygen to form nitrogen dioxide under discharge conditions)

答案 (Answer): D

解析 (Analysis): 氮在自然界中存在游离态, 如空气中的 N_2 , 也有化合态, 如蛋白质中的氮, A 项正确; 氮气分子内存在氮氮三键, 键能很高, 氮气化学性质稳定, 一般不和其他物质发生反应, B 项正确; 在反应

$3\text{Mg} + \text{N}_2 \xrightarrow{\text{点燃}} \text{Mg}_3\text{N}_2$ 中, N 元素化合价降低, N_2 作氧化剂, C 项正确; 放电条件下氮气与氧气反应生成一氧化氮, D 项错误。 (Nitrogen exists in free state in nature, such as N_2 in the air, and also in combined state, such as nitrogen in proteins. Item A is correct. There is a nitrogen-nitrogen triple bond in nitrogen molecules with high bond energy, so nitrogen gas is chemically stable and generally does not react with other substances. Item B is correct. In the reaction $3\text{Mg} + \text{N}_2 \xrightarrow{\text{ignition}} \text{Mg}_3\text{N}_2$, the valence of nitrogen element decreases and N_2 acts as an oxidizing agent. Item C is correct. Nitrogen gas reacts with oxygen to form nitric oxide under discharge conditions. Item D is incorrect.)

2. 按图进行 NO_2 还原性的探究实验。随着实验的进行, 下列说法正确的是 (Conduct an experiment to explore the reducing property of NO_2 according to the figure. Which of the following statements is correct as the experiment proceeds?) ()



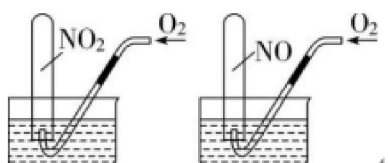
- A. 可观察到装置 I 中逐渐产生红棕色气体 (Reddish-brown gas can be observed gradually produced in Device I)
B. 装置 II 中的试剂可以更换为碱石灰 (The reagent in Device II can be replaced with soda lime)
C. 可观察到装置 III 中的气体和固体颜色变浅 (The color of gas and solid in Device III can be observed to become lighter)
D. 装置 IV 中发生反应的离子方程式为 $\text{NO}_2 + 2\text{OH}^- = \text{NO}_3^- + \text{NO}_2^- + \text{H}_2\text{O}$ (The ionic equation of the reaction in Device IV is $\text{NO}_2 + 2\text{OH}^- = \text{NO}_3^- + \text{NO}_2^- + \text{H}_2\text{O}$)

答案 (Answer): C

解析 (Analysis): 浓硝酸和铜反应生成硝酸铜、二氧化氮和水, 经过无水氯化钙干燥得到较纯净的 NO_2 , NO_2 在 III 中被过氧化钠氧化后用氢氧化钠溶液作尾气处理。铜与浓硝酸反应迅速生成 NO_2 , 因此加入浓硝酸后会立即看到红棕色气体, A 错误; 碱石灰可以吸收 NO_2 , 会干扰 NO_2 与过氧化钠的反应, B 错误; 装置 III 中发生反应 $2\text{NO}_2 + \text{Na}_2\text{O}_2 = 2\text{NaNO}_3$, 故 NO_2 的红棕色、 Na_2O_2 的淡黄色均会变浅, C 正确; 装置 IV 中发生反应的离子方程式为 $2\text{NO}_2 + 2\text{OH}^- = \text{NO}_3^- + \text{NO}_2^- + \text{H}_2\text{O}$, D 错误。 (Concentrated nitric acid reacts with copper to produce copper nitrate, nitrogen dioxide and water. Pure NO_2 is obtained after drying with anhydrous calcium chloride. NO_2 is oxidized by sodium peroxide in Device III, and then sodium hydroxide solution is used for tail gas treatment. Copper reacts rapidly with concentrated nitric acid to form NO_2 , so reddish-brown gas can be seen immediately after adding concentrated nitric acid. A is wrong. Soda lime can absorb NO_2 and interfere with the reaction between NO_2 and sodium peroxide. B is wrong. The reaction $2\text{NO}_2 + \text{Na}_2\text{O}_2 = 2\text{NaNO}_3$ occurs in Device III, so the reddish-brown color of NO_2 and the light yellow color of Na_2O_2 both become lighter. C is

correct. The ionic equation of the reaction in Device IV is $2\text{NO}_2 + 2\text{OH}^- = \text{NO}_3^- + \text{NO}_2^- + \text{H}_2\text{O}$. D is wrong.)

3. 如图所示, 相同条件下, 两个容积相同的试管分别装满 NO_2 和 NO 气体, 分别倒置于水槽中, 然后通过导管缓慢通入氧气, 边通 O_2 边慢慢摇动试管, 直到两个试管内充满液体。假设试管内的溶质不向水槽中扩散, 则两个试管内溶液物质的量浓度之比为 (As shown in the figure, under the same conditions, two test tubes with the same volume are filled with NO_2 and NO gas respectively, inverted into a water tank, and then oxygen is slowly introduced through a catheter. Shake the test tubes slowly while introducing O_2 until both test tubes are filled with liquid. Assuming that the solute in the test tubes does not diffuse into the water tank, what is the ratio of the molar concentration of the solution in the two test tubes?) ()

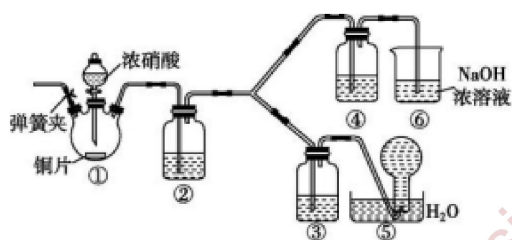


- A. 1 : 1
B. 5 : 7
C. 7 : 5
D. 4 : 3

答案 (Answer): A

解析 (Analysis): 相同条件下, 用 NO_2 、 NO 装满试管, 则体积相同, 根据阿伏伽德罗定律: $n(\text{NO}_2) = n(\text{NO})$, 根据氮原子守恒可得试管内硝酸的物质的量相同, 溶液的体积相同, 所以两个试管中溶液物质的量浓度之比为1 : 1。
(Under the same conditions, filling the test tubes with NO_2 and NO results in the same volume. According to Avogadro's law: $n(\text{NO}_2) = n(\text{NO})$. According to the conservation of nitrogen atoms, the amount of nitric acid in the test tubes is the same, and the volume of the solution is the same. Therefore, the ratio of the molar concentration of the solution in the two test tubes is 1 : 1.)

4. 资料显示: 浓硝酸能将 NO 氧化成 NO_2 , 而稀硝酸不能氧化 NO 。某学习小组据此验证浓、稀硝酸氧化性的相对强弱, 按如图装置进行实验 (夹持仪器已略去)。可选试剂: 浓硝酸、稀硝酸 ($3\text{ mol} \cdot \text{L}^{-1}$)、蒸馏水、浓硫酸、 NaOH 溶液。下列说法错误的是 (Data shows that concentrated nitric acid can oxidize NO to NO_2 , while dilute nitric acid cannot oxidize NO . A study group verified the relative strength of oxidizing property of concentrated and dilute nitric acid according to this, and conducted the experiment according to the device shown in the figure (clamping instruments are omitted). Optional reagents: concentrated nitric acid, dilute nitric acid ($3\text{ mol} \cdot \text{L}^{-1}$), distilled water, concentrated sulfuric acid, NaOH solution. Which of the following statements is incorrect?) ()



- A. 实验开始前及结束后均需通一段时间氮气, 其目的不同 (Nitrogen gas needs to be introduced for a period of time before and after the experiment, and the purposes are different)
B. 装置②盛放的试剂为蒸馏水 (The reagent contained in Device ② is distilled water)
C. 做实验时应避免将有害气体排放到空气中, 故装置③④盛放的试剂依次是浓硝酸、稀硝酸 (Harmful gases should be prevented from being discharged into the air during the experiment, so the reagents contained in Device ③ and ④ are concentrated nitric acid and dilute nitric acid in turn)
D. 能证明氧化性的相对强弱的实验现象是装置③中气体仍为无色而装置④中气体变为红棕色 (The experimental phenomenon that can prove the relative strength of oxidizing property is that the gas in Device ③ remains colorless while that in Device ④ turns reddish-brown)

答案 (Answer): C

解析 (Analysis): 装置①中浓硝酸与铜反应生成硝酸铜、 NO_2 和水, 通过盛有蒸馏水的装置②, NO_2 与水反应生成 NO , NO 气体通过盛有稀硝酸的装置③, 不发生反应, 最后用排水法在装置⑤中收集到 NO 气体; NO 气体通过盛有浓硝酸的装置④, 浓硝酸能把 NO 氧化为 NO_2 , 装置④中气体为红棕色, 尾气被盛有 NaOH 溶液的装置⑥吸收。实验开始前需通一段时间氮气, 目的是排出装置中的氧气, 结束后通氮气目的是使装置中残留的氮氧化物被收集或者吸收, 防

止污染空气，二者目的不同，A 正确；装置②盛放的试剂为蒸馏水， NO_2 与水反应生成 NO ，B 正确；装置③④盛放的试剂依次是稀硝酸、浓硝酸，用来验证浓硝酸能将 NO 氧化成 NO_2 ，而稀硝酸不能氧化 NO ，为避免将有害气体排放到空气中，装置⑥盛放的 NaOH 溶液吸收 NO_2 ，装置⑤收集 NO ，C 错误；根据实验现象：③中气体仍为无色而④中气体变为红棕色，说明浓硝酸能把 NO 氧化为 NO_2 ，稀硝酸不能氧化，可得出浓硝酸氧化性强于稀硝酸，D 正确。

(Concentrated nitric acid reacts with copper in Device ① to produce copper nitrate, NO_2 and water. NO_2 reacts with water to form NO through Device ② containing distilled water. NO gas passes through Device ③ containing dilute nitric acid without reaction, and finally NO gas is collected in Device ⑤ by water displacement method. NO gas passes through Device ④ containing concentrated nitric acid, and concentrated nitric acid can oxidize NO to NO_2 . The gas in Device ④ is reddish-brown, and the tail gas is absorbed by Device ⑥ containing NaOH solution. Nitrogen gas needs to be introduced for a period of time before the experiment to expel oxygen in the device, and after the experiment to collect or absorb residual nitrogen oxides in the device to prevent air pollution. The two purposes are different. A is correct. The reagent in Device ② is distilled water, and NO_2 reacts with water to form NO . B is correct. The reagents in Device ③ and ④ are dilute nitric acid and concentrated nitric acid in turn to verify that concentrated nitric acid can oxidize NO to NO_2 while dilute nitric acid cannot. To prevent harmful gases from being discharged into the air, Device ⑥ containing NaOH solution absorbs NO_2 and Device ⑤ collects NO . C is wrong. According to the experimental phenomenon: the gas in ③ remains colorless while that in ④ turns reddish-brown, indicating that concentrated nitric acid can oxidize NO to NO_2 and dilute nitric acid cannot. It can be concluded that the oxidizing property of concentrated nitric acid is stronger than that of dilute nitric acid. D is correct.)

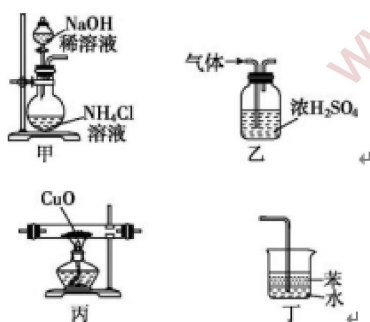
5. 下列有关氨和铵盐的叙述正确的是 (Which of the following statements about ammonia and ammonium salts is correct?) ()

- A. 液氨常用作制冷剂 (Liquid ammonia is often used as a refrigerant)
- B. NH_4Cl 中含少量的 I_2 ，可通过加热的方法除去 (A small amount of I_2 in NH_4Cl can be removed by heating)
- C. 将蘸有浓氨水和浓硫酸的玻璃棒相互靠近，有白烟产生 (White smoke is produced when glass rods dipped in concentrated ammonia water and concentrated sulfuric acid are close to each other)
- D. 某气体通入紫色石蕊溶液后溶液变红，据此可推断该气体为 NH_3 (A gas turns purple litmus solution red, so it can be inferred that the gas is NH_3)

答案 (Answer): A

解析 (Analysis): 液氨汽化时吸收大量的热导致周围温度急剧降低，所以液氨常用作制冷剂，故 A 正确；加热氯化铵会分解为 NH_3 和 HCl ，遇冷又生成 NH_4Cl ，加热时 I_2 会升华，所以不能用加热的方法除去 NH_4Cl 中的少量 I_2 ，故 B 错误；将蘸有浓氨水和挥发性酸（如盐酸和硝酸）的玻璃棒相互靠近，有白烟产生，而硫酸不是挥发性的酸，故 C 错误；氨通入紫色石蕊溶液后溶液应变蓝，而不是变红，故 D 错误。(Liquid ammonia absorbs a lot of heat during vaporization, causing the surrounding temperature to drop sharply, so liquid ammonia is often used as a refrigerant. A is correct. Heating ammonium chloride decomposes into NH_3 and HCl , which regenerate NH_4Cl when cooled. I_2 sublimates when heated, so a small amount of I_2 in NH_4Cl cannot be removed by heating. B is wrong. White smoke is produced when glass rods dipped in concentrated ammonia water and volatile acids (such as hydrochloric acid and nitric acid) are close to each other, while sulfuric acid is not a volatile acid. C is wrong. Ammonia turns purple litmus solution blue instead of red. D is wrong.)

6. 下列装置用于实验室制氨气或验证氨气的化学性质，其中能达到实验目的的是 (The following devices are used to prepare ammonia or verify the chemical properties of ammonia in the laboratory. Which one can achieve the experimental purpose?) ()



- A. 用装置甲制取氨气 (Prepare ammonia with Device Jia)
- B. 用装置乙除去氨气中的水蒸气 (Remove water vapor in ammonia with Device Yi)
- C. 用装置丙验证氨气具有还原性 (Verify the reducing property of ammonia with Device Bing)
- D. 用装置丁吸收尾气 (Absorb tail gas with Device Ding)

答案 (Answer):C

解析 (Analysis): NH_3 极易溶于水, NaOH 稀溶液与 NH_4Cl 溶液混合生成 NaCl 和 $\text{NH}_3 \cdot \text{H}_2\text{O}$, 不易释放出 NH_3 , A 错误; NH_3 可被浓硫酸吸收生成 $(\text{NH}_4)_2\text{SO}_4$, B 错误; NH_3 与 CuO 在加热条件下反应, 黑色固体变成红色, CuO 被还原为 Cu , 体现 NH_3 的还原性, C 正确; NH_3 极易溶于水, 导管末端插入水中起不到防倒吸作用, D 错误。 (NH_3 is extremely soluble in water. Mixing dilute NaOH solution with NH_4Cl solution produces NaCl and $\text{NH}_3 \cdot \text{H}_2\text{O}$, which is not easy to release NH_3 . A is wrong. NH_3 can be absorbed by concentrated sulfuric acid to form $(\text{NH}_4)_2\text{SO}_4$. B is wrong. NH_3 reacts with CuO under heating conditions, the black solid turns red, and CuO is reduced to Cu , reflecting the reducing property of NH_3 . C is correct. NH_3 is extremely soluble in water, and inserting the end of the catheter into water cannot prevent back suction. D is wrong.)

7. 用充有 NH_3 的烧瓶进行“喷泉实验”, 实验装置及现象如图所示。下列关于该实验的分析正确的是 (Conduct a fountain experiment with a flask filled with NH_3 . The experimental device and phenomena are shown in the figure. Which of the following analyses of the experiment is correct?) ()

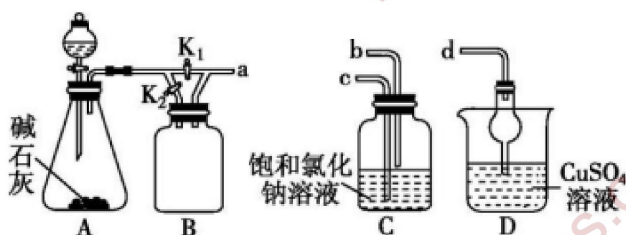


- A. 产生“喷泉”证明 NH_3 与 H_2O 发生了反应 (The formation of “fountain” proves that NH_3 reacts with H_2O)
- B. 无色溶液变红证明 NH_3 极易溶于水 (The colorless solution turns red, proving that NH_3 is extremely soluble in water)
- C. 红色溶液中存在含有离子键的含氮微粒 (Nitrogen-containing particles with ionic bonds exist in the red solution)
- D. 加热红色溶液可观察到红色变浅或褪去 (Heating the red solution can observe the red color becomes lighter or fades)

答案 (Answer):D

解析 (Analysis): 该实验利用氨气极易溶于水生成碱性溶液从而产生压强差, 出现喷泉现象。喷泉的形成说明氨气极易溶于水从而产生了压强差, 不能说明两者反应了, A 错误; 无色酚酞溶液遇碱变为红色, 氨气溶于水后溶液由无色变为红色, 说明氨水呈碱性, B 错误; 红色溶液中存在的含氮微粒为铵根离子、一水合氨, 两者均含有共价键、不含有离子键, C 错误; 氨水受热不稳定, 易分解放出氨气, 从而使溶液的碱性减弱, 故滴有酚酞且呈红色的氨水受热后红色变浅或褪色, D 正确。 (This experiment uses the fact that ammonia is extremely soluble in water to form an alkaline solution, resulting in a pressure difference and the fountain phenomenon. The formation of the fountain indicates that ammonia is extremely soluble in water, resulting in a pressure difference, but does not indicate that the two react. A is wrong. Colorless phenolphthalein solution turns red in alkaline solution. After ammonia dissolves in water, the solution turns from colorless to red, indicating that ammonia water is alkaline. B is wrong. The nitrogen-containing particles in the red solution are ammonium ions and ammonia monohydrate, both of which contain covalent bonds but no ionic bonds. C is wrong. Ammonia water is unstable when heated and easily decomposes to release ammonia, weakening the alkalinity of the solution. Therefore, the red ammonia water with phenolphthalein turns lighter or fades when heated. D is correct.)

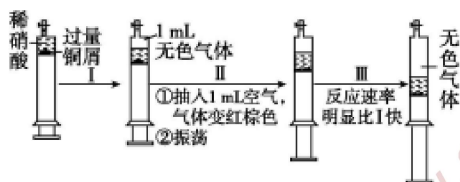
8. 利用如图所示装置进行 NH_3 的制备及性质探究实验。下列说法正确的是 (Conduct an experiment on the preparation and property exploration of NH_3 with the device shown in the figure. Which of the following statements is correct?) ()



- A. 使用 A 装置制备 NH_3 时, 分液漏斗中应盛放浓氨水, 利用了氨水的碱性弱于 NaOH (When preparing NH_3 with Device A, concentrated ammonia water should be placed in the separatory funnel, utilizing that the alkalinity of ammonia water is weaker than that of NaOH)
- B. 装置 B 中盛放湿润蓝色石蕊试纸, 打开 K_2 , 关闭 K_1 , 可证明 NH_3 为碱性气体 (Put moist blue litmus paper in



10.一定温度下，探究铜与稀硝酸反应的过程如下：(At a certain temperature, the process of the reaction between copper and dilute nitric acid is explored as follows:)



下列说法不正确的是 () (Which of the following statements is incorrect? ())

- A. 过程 I 中被还原的硝酸占反应硝酸总物质的量的 $\frac{1}{4}$ (In Process I, the nitric acid that is reduced accounts for $\frac{1}{4}$ of the total amount of reacted nitric acid)
- B. 过程 III 的反应速率比过程 I 快的原因是 NO_2 溶于水, 使 HNO_3 浓度增大 (The reason why the reaction rate of Process III is faster than that of Process I is that NO_2 dissolves in water, increasing the concentration of HNO_3)
- C. 由实验可知, NO_2 对该反应可能具有催化作用 (It can be seen from the experiment that NO_2 may have a catalytic effect on this reaction)
- D. 当活塞不再移动时, 再抽入空气, 铜可以继续溶解 (When the piston stops moving, air is drawn in again, and copper can continue to dissolve)

答案 (Answer): B

解析 (Analysis): 过程 I 中反应为 $3\text{Cu} + 8\text{HNO}_3(\text{稀}) = 3\text{Cu}(\text{NO}_3)_2 + 2\text{NO} \uparrow + 4\text{H}_2\text{O}$, 硝酸被还原的产物为 NO , 通过反应方程式可得出被还原的硝酸占反应硝酸总物质的量的 $\frac{1}{4}$, A 正确; 二氧化氮与水反应生成硝酸和 NO , 化学方程式为 $3\text{NO}_2 + \text{H}_2\text{O} = 2\text{HNO}_3 + \text{NO}$, 由氮原子守恒可知, 过程 III 中硝酸的浓度比过程 I 小, 因为部分硝酸转化成 NO , 因此过程 III 反应速率比过程 I 快的原因不可能是 $c(\text{HNO}_3)$ 增大, B 错误; 由 B 项分析可知, $c(\text{HNO}_3)$: I > III, 但过程 II 生成 NO_2 , NO_2 可能对铜与稀硝酸反应具有催化作用, 使反应速率加快, C 正确; 当活塞不再移动时, 再抽入空气, NO 与 O_2 反应生成 NO_2 , NO_2 溶于水生成稀硝酸, 能够继续溶解铜, D 正确。 (In Process I, the reaction is $3\text{Cu} + 8\text{HNO}_3(\text{dilute}) = 3\text{Cu}(\text{NO}_3)_2 + 2\text{NO} \uparrow + 4\text{H}_2\text{O}$. The reduction product of nitric acid is NO . It can be concluded from the reaction equation that the nitric acid that is reduced accounts for $\frac{1}{4}$ of the total amount of reacted nitric acid, so A is correct. Nitrogen dioxide reacts with water to generate nitric acid and NO , and the chemical equation is $3\text{NO}_2 + \text{H}_2\text{O} = 2\text{HNO}_3 + \text{NO}$. According to the conservation of nitrogen atoms, the concentration of nitric acid in Process III is lower than that in Process I because part of the nitric acid is converted into NO . Therefore, the faster reaction rate of Process III cannot be due to the increase in $c(\text{HNO}_3)$, so B is incorrect. According to the analysis of Item B, $c(\text{HNO}_3)$: I > III, but NO_2 is generated in Process II. NO_2 may have a catalytic effect on the reaction between copper and dilute nitric acid, accelerating the reaction rate, so C is correct. When the piston stops moving, air is drawn in again. NO reacts with O_2 to generate NO_2 , and NO_2 dissolves in water to form dilute nitric acid, which can continue to dissolve copper, so D is correct.)

11. 对比分析下列实验, 所得结论正确的是 () (The following conclusions obtained by comparative analysis of the following experiments are correct? ())



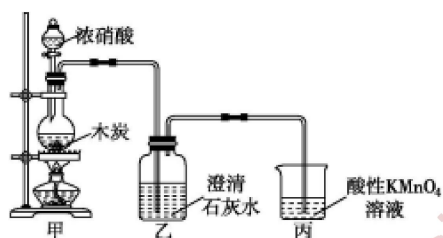
- A. 由①中收集的气体能使带火星木条复燃, 可推断 NO_2 能助燃 (The gas collected in ① can reignite a glowing splint, so it can be inferred that NO_2 supports combustion)
- B. ②中产生红棕色气体可以证明浓硝酸有不稳定性 (The reddish-brown gas generated in ② can prove that concentrated nitric acid is unstable)
- C. ③中气体产物中有 CO_2 , 能证明 C 能将硝酸还原 (The gas product in ③ contains CO_2 , which can prove that C can reduce nitric acid)
- D. ①②③中产生的气体通入足量水中后都无污染性气体产生 (After the gases produced in ①, ② and ③ are passed into sufficient water, no pollutant gas is generated)

答案 (Answer): A

解析 (Analysis): ①中发生反应 $4\text{HNO}_3(\text{浓}) = 4\text{NO}_2 \uparrow + \text{O}_2 \uparrow + 2\text{H}_2\text{O}$, 收集的气体中 O_2 体积的百分含量为 20%, 与空气中 O_2 体积的百分含量 (约 21%) 接近, 而空气不能使带火星的木条复燃, 由此可推断 NO_2 能助燃, A 正确; 浓硝酸能与红热的木炭反应生成 NO_2 , 不能证明浓硝酸具有不稳定性, B 错误; 红热的木炭与试管中的氧气反应也能生成二氧化碳, C 错误; ②③中生成的 NO_2 溶于水后生成 HNO_3 和 NO , NO 属于污染性气体, D 错误。 (The

reaction in ① is $4\text{HNO}_3(\text{concentrated}) = 4\text{NO}_2 \uparrow + \text{O}_2 \uparrow + 2\text{H}_2\text{O}$. The volume percentage of O_2 in the collected gas is 20%, which is close to that in air (about 21%), but air fails to reignite a glowing splint. Thus it can be inferred that NO_2 supports combustion, so A is correct. Concentrated nitric acid can react with red-hot charcoal to generate NO_2 , which cannot prove that concentrated nitric acid is unstable, so B is incorrect. Red-hot charcoal can also react with oxygen in the test tube to generate carbon dioxide, so C is incorrect. NO_2 generated in ② and ③ dissolves in water to form HNO_3 and NO , and NO is a pollutant gas, so D is incorrect.)

12. 某实验小组用如图所示装置探究木炭与浓硝酸的反应，实验记录如下：(An experimental group explores the reaction between charcoal and concentrated nitric acid with the device shown in the figure, and the experimental records are as follows:)



①烧瓶中产生红棕色气体；②澄清石灰水始终不变浑浊；③酸性 KMnO_4 溶液颜色变浅。(① Reddish-brown gas is generated in the flask; ② The clear lime water never turns turbid; ③ The color of acidic KMnO_4 solution fades.)

下列叙述正确的是 () (Which of the following statements is correct? ())

- A. 甲中木炭一定与浓硝酸发生了反应 (Charcoal in Flask A must react with concentrated nitric acid)
- B. 乙向丙中导入了还原性气体 (Device B introduces a reducing gas into Device C)
- C. 乙中不变浑浊的原因是甲中不产生 CO_2 (The reason why Device B does not turn turbid is that no CO_2 is generated in Flask A)
- D. 浓硝酸与木炭的反应中，硝酸表现氧化性和酸性 (In the reaction between concentrated nitric acid and charcoal, nitric acid shows oxidizing property and acidity)

答案 (Answer): B

解析 (Analysis): 浓硝酸受热易分解生成 NO_2 和 O_2 ，根据现象不能证明甲中木炭与浓硝酸发生了反应，A 错误； KMnO_4 是氧化剂，由“酸性 KMnO_4 溶液颜色变浅”推知：导入丙中的气体含有还原性气体成分，如 NO ，B 正确；如果甲中产生 CO_2 和大量 NO_2 ， NO_2 和 H_2O 反应生成 HNO_3 和 NO ， HNO_3 溶解了 CaCO_3 ，则乙中不会变浑浊，C 错误；浓硝酸与木炭反应生成 NO_2 、 CO_2 和 H_2O ，硝酸只表现氧化性，D 错误。(Concentrated nitric acid is easily decomposed by heat to generate NO_2 and O_2 . The phenomenon cannot prove that charcoal in Flask A reacts with concentrated nitric acid, so A is incorrect. KMnO_4 is an oxidizing agent. It can be inferred from the color of acidic KMnO_4 solution fades that the gas introduced into Device C contains reducing gas components such as NO , so B is correct. If CO_2 and a large amount of NO_2 are generated in Flask A, NO_2 reacts with H_2O to form HNO_3 and NO , and HNO_3 dissolves CaCO_3 , so Device B will not turn turbid, so C is incorrect. Concentrated nitric acid reacts with charcoal to generate NO_2 , CO_2 and H_2O , and nitric acid only shows oxidizing property, so D is incorrect.)

13. 足量铜与一定量的浓硝酸反应，得到硝酸铜溶液和 NO_2 、 NO 的混合气体 2.24 L，这些气体与一定体积 O_2 混合后通入水中，所有气体完全被水吸收生成硝酸。若向所得硝酸铜溶液中加入 $4 \text{ mol} \cdot \text{L}^{-1}$ NaOH 溶液至 Cu^{2+} 恰好完全沉淀，则消耗 NaOH 溶液的体积是 50 mL，下列说法正确的是 (气体体积均已折算为标准状况，下同)() (Sufficient copper reacts with a certain amount of concentrated nitric acid to obtain a copper nitrate solution and 2.24 L of a mixed gas of NO_2 and NO . These gases are mixed with a certain volume of O_2 and passed into water, and all gases are completely absorbed by water to form nitric acid. If $4 \text{ mol} \cdot \text{L}^{-1}$ NaOH solution is added to the obtained copper nitrate solution until Cu^{2+} is completely precipitated, the volume of consumed NaOH solution is 50 mL. Which of the following statements is correct (all gas volumes are converted to standard conditions, the same below)? ())

- A. 参加反应的 HNO_3 的物质的量为 0.2 mol (The amount of HNO_3 participating in the reaction is 0.2 mol)
- B. 消耗氧气的体积为 2.24 L (The volume of consumed oxygen is 2.24 L)
- C. 铜和硝酸反应过程中转移电子的物质的量为 0.3 mol (The amount of transferred electrons in the reaction between copper and nitric acid is 0.3 mol)
- D. 混合气体中含 1.12 L NO_2 (The mixed gas contains 1.12 L of NO_2)

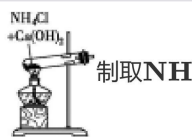
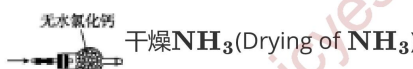
答案 (Answer): D

解析 (Analysis): 沉淀量达到最大，溶液中溶质为 NaNO_3 ，根据 N 原子守恒，参加反应的 HNO_3 的物质的量为 $n(\text{NaNO}_3) = n(\text{NO}) + n(\text{NO}_2) = (50 \times 10^{-3} \times 4 + \frac{2.24}{22.4}) \text{ mol} = 0.3 \text{ mol}$ ，故 A 项错误；



$\text{Cu}^{2+} + 2\text{OH}^- = \text{Cu}(\text{OH})_2 \downarrow$, 根据离子方程式求出参加反应的Cu的物质的量为
 $n(\text{Cu}) = \frac{1}{2} \times (50 \times 10^{-3} \times 4) \text{ mol} = 0.1 \text{ mol}$, 根据整个实验, Cu失去的电子最终给了氧气, 根据得失电子数目守恒, 即 $n(\text{Cu}) \times 2 = n(\text{O}_2) \times 4$, 则 $n(\text{O}_2) = \frac{1}{2} \times 0.1 \text{ mol} = 0.05 \text{ mol}$, 在标准状况下氧气的体积为
 $0.05 \text{ mol} \times 22.4 \text{ L} \cdot \text{mol}^{-1} = 1.12 \text{ L}$, 故 B 项错误; 铜和硝酸反应, 转移电子的物质的量为 $0.1 \text{ mol} \times 2 = 0.2 \text{ mol}$, 故 C 项错误; 设NO的物质的量为 $x \text{ mol}$, 则 NO_2 的物质的量为 $(0.1 - x) \text{ mol}$, 根据得失电子数目守恒, 即
 $3x \text{ mol} + (0.1 - x) \text{ mol} = 0.2 \text{ mol}$, 解得 $x = 0.05$, 因此 NO_2 的体积 (折算为标准状况) 为
 $(0.1 - 0.05) \text{ mol} \times 22.4 \text{ L} \cdot \text{mol}^{-1} = 1.12 \text{ L}$, 故 D 项正确。 (When the precipitation reaches the maximum, the solute in the solution is NaNO_3 . According to the conservation of nitrogen atoms, the amount of HNO_3 participating in the reaction is $n(\text{NaNO}_3) + n(\text{NO}) + n(\text{NO}_2) = (50 \times 10^{-3} \times 4 + \frac{2.24}{22.4}) \text{ mol} = 0.3 \text{ mol}$, so Item A is incorrect. $\text{Cu}^{2+} + 2\text{OH}^- = \text{Cu}(\text{OH})_2 \downarrow$. According to the ionic equation, the amount of Cu participating in the reaction is $n(\text{Cu}) = \frac{1}{2} \times (50 \times 10^{-3} \times 4) \text{ mol} = 0.1 \text{ mol}$. According to the whole experiment, the electrons lost by Cu are finally transferred to oxygen. According to the conservation of the number of gained and lost electrons, $n(\text{Cu}) \times 2 = n(\text{O}_2) \times 4$, then $n(\text{O}_2) = \frac{1}{2} \times 0.1 \text{ mol} = 0.05 \text{ mol}$. The volume of oxygen under standard conditions is $0.05 \text{ mol} \times 22.4 \text{ L} \cdot \text{mol}^{-1} = 1.12 \text{ L}$, so Item B is incorrect. In the reaction between copper and nitric acid, the amount of transferred electrons is $0.1 \text{ mol} \times 2 = 0.2 \text{ mol}$, so Item C is incorrect. Let the amount of NO be $x \text{ mol}$, then the amount of NO_2 is $(0.1 - x) \text{ mol}$. According to the conservation of the number of gained and lost electrons, $3x \text{ mol} + (0.1 - x) \text{ mol} = 0.2 \text{ mol}$, and $x = 0.05$ is obtained. Therefore, the volume of NO_2 (converted to standard conditions) is $(0.1 - 0.05) \text{ mol} \times 22.4 \text{ L} \cdot \text{mol}^{-1} = 1.12 \text{ L}$, so Item D is correct.)

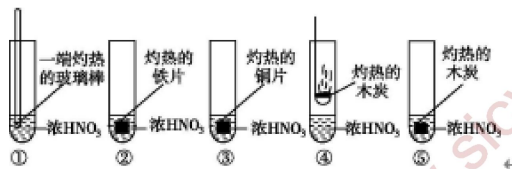
14. NH_3 是一种广泛应用于工业、农业和医药领域的重要化工气体, 下列有关氨气的实验中, 能达到实验目的的是 () (NH_3 is an important chemical gas widely used in industry, agriculture and medicine. Which of the following experiments related to ammonia can achieve the experimental purpose? ())

A	B
 制取 NH_3 (Preparation of NH_3)	 干燥 NH_3 (Drying of NH_3)
C	D
 收集 NH_3 (Collection of NH_3)	 先从 a 口通 NH_3, 再从 b 口通 CO_2 制 NaHCO_3 (First pass NH_3 through port a, then pass CO_2 through port b to prepare NaHCO_3)

答案 (Answer): D

解析 (Analysis): 加热固体制备气体时, 试管口应略向下倾斜, A 错误; 不能用无水氯化钙干燥 NH_3 , B 错误; NH_3 密度比空气小, 应用向下排空气法收集, C 错误; CO_2 在水中溶解度较小, 制备 NaHCO_3 时先从 a 口通 NH_3 (可防倒吸), 再从 b 口通 CO_2 , D 正确。 (When preparing gas by heating solid, the test tube mouth should be slightly inclined downward, so A is incorrect. Anhydrous calcium chloride cannot be used to dry NH_3 , so B is incorrect. The density of NH_3 is lower than that of air, so it should be collected by downward air displacement method, so C is incorrect. The solubility of CO_2 in water is small. When preparing NaHCO_3 , first pass NH_3 through port a (to prevent back suction), then pass CO_2 through port b, so D is correct.)

15. 为了探究浓硝酸的性质, 进行以下五组实验, 均观察到红棕色气体。下列分析错误的是 () (To explore the properties of concentrated nitric acid, the following five groups of experiments are carried out, and reddish-brown gas is observed in all of them. Which of the following analyses is incorrect? ())



A. 将带火星的木条靠近①的试管口，木条复燃，说明 NO_2 能支持燃烧 (Put a glowing splint near the mouth of test tube ①, and the splint reignites, indicating that NO_2 can support combustion)

B. ②中的铁片钝化，红棕色气体由浓硝酸分解产生，③中溶液变为绿色 (The iron sheet in ② is passivated, the reddish-brown gas is generated by the decomposition of concentrated nitric acid, and the solution in ③ turns green)

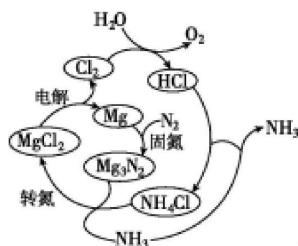
C. ①④⑤实验对比, ④中的红棕色气体可能是由挥发的浓硝酸受热分解产生的 (Compared with experiments ①, ④ and ⑤, the reddish-brown gas in ④ may be generated by the thermal decomposition of volatile concentrated nitric acid)

D. 五支试管中的红棕色气体均为还原产物 (The reddish-brown gas in the five test tubes is all reduction product)

答案 (Answer): B

解析 (Analysis): 灼热的玻璃棒插入浓硝酸中, HNO_3 分解生成 NO_2 、 O_2 和 H_2O , 此时 O_2 的含量与空气中 O_2 的含量相当, 带火星的木条靠近①的试管口, 木条复燃, 说明 NO_2 能支持燃烧, A 正确; 灼热的铁片在浓硝酸中发生反应, 可生成红棕色的 NO_2 , 不发生钝化, B 错误; 浓硝酸具有强挥发性, 挥发出的 $\text{HNO}_3(\text{g})$ 受灼热木炭作用, 可发生分解反应生成红棕色的 NO_2 , C 正确; 红棕色气体均为 NO_2 , 都是由 HNO_3 发生还原反应生成的, 均为还原产物, D 正确。(A hot glass rod is inserted into concentrated nitric acid, and HNO_3 decomposes to generate NO_2 , O_2 and H_2O . At this time, the content of O_2 is equivalent to that in air. A glowing splint is put near the mouth of test tube ① and reignites, indicating that NO_2 can support combustion, so A is correct. The hot iron sheet reacts in concentrated nitric acid to generate reddish-brown NO_2 without passivation, so B is incorrect. Concentrated nitric acid has strong volatility. The volatilized $\text{HNO}_3(\text{g})$ is acted by hot charcoal and can decompose to generate reddish-brown NO_2 , so C is correct. The reddish-brown gas is all NO_2 , which is generated by the reduction reaction of HNO_3 and is all reduction product, so D is correct.)

16. 一种新型合成氨的过程如图所示。下列说法正确的是 () (A new process for ammonia synthesis is shown in the figure. Which of the following statements is correct? ())



A. NH_3 和 NH_4^+ 的中心原子杂化方式相同 (The central atom hybridization modes of NH_3 and NH_4^+ are the same)

B. “固氮”和“转氮”过程均属于人工固氮 (Both the processes of nitrogen fixation and nitrogen conversion belong to artificial nitrogen fixation)

C. 该合成氨过程中, 参加反应的 N_2 与 H_2O 的物质的量之比为 3 : 1 (In this ammonia synthesis process, the ratio of the amount of N_2 to H_2O participating in the reaction is 3:1)

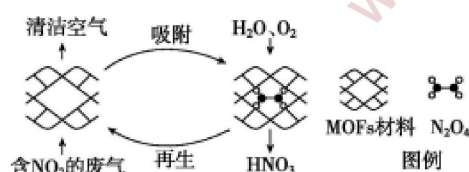
D. 该合成氨过程中，所有反应均为氧化还原反应 (In this ammonia synthesis process, all reactions are redox reactions)

答案 (Answer): A

解析 (Analysis): NH_3 和 NH_4^+ 的中心原子N的价层电子对数分别为 $3 + \frac{1}{2} \times (5 - 3 \times 1) = 4$, $4 + \frac{1}{2} \times (5 - 1 - 4 \times 1) = 4$, 故中心原子均为 sp^3 杂化, A 正确; 由游离态的氮转化为化合态的氮为氮的固定, 故“转氮”过程没有氮气参与, 不属于人工固氮, B 错误; 合成氨过程中, N_2 最终转化为 NH_3 , 1 mol N_2 转移 6 mol 电子, H_2O 转化为 O_2 , 1 mol H_2O 转移 2 mol 电子, 由得失电子守恒可知, 参加反应的 N_2 与 H_2O 的物质的量之比为 1 : 3, C 错误; 不是所有反应都是氧化还原反应, 如生成 NH_4Cl 的过程, D 错误。 (The valence electron pairs of the central atom N of NH_3 and NH_4^+ are $3 + \frac{1}{2} \times (5 - 3 \times 1) = 4$ and $4 + \frac{1}{2} \times (5 - 1 - 4 \times 1) = 4$ respectively, so the central atoms are both sp^3 hybridized, so A is correct. The conversion of free nitrogen to combined nitrogen is nitrogen fixation. Therefore, the nitrogen conversion process does not involve nitrogen and does not belong to artificial nitrogen fixation, so B is incorrect. In the ammonia synthesis process, N_2 is finally converted to NH_3 , 1 mol of N_2 transfers 6 mol of electrons, H_2O is converted to O_2 , and 1 mol of H_2O transfers 2 mol of electrons. According to the conservation of gained and lost electrons, the ratio of the amount

of N_2 to H_2O participating in the reaction is 1:3, so C is incorrect. Not all reactions are redox reactions, such as the process of forming NH_4Cl , so D is incorrect.)

17. 某 MOFs 多孔材料孔径大小和形状恰好将 N_2O_4 “固定”，利于工业废气中的 NO_2 高效转化为 HNO_3 ，其原理示意图如下。(A certain MOFs porous material just fixes N_2O_4 with its pore size and shape, which is conducive to the efficient conversion of NO_2 in industrial waste gas into HNO_3 . The schematic diagram of its principle is as follows.)



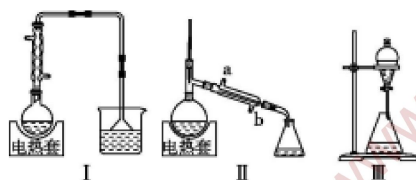
已知: $2NO_2(g) \rightleftharpoons N_2O_4(g)$ $\Delta H < 0$. 下列说法正确的是 () (Given: $2NO_2(g) \rightleftharpoons N_2O_4(g)$ $\Delta H < 0$. Which of the following statements is correct? ())

- A. 该过程属于氮的固定 (This process belongs to nitrogen fixation)
- B. “固定”过程体现超分子的分子识别功能 (The “fixation” process reflects the molecular recognition function of supramolecules)
- C. 温度升高时有利于 N_2O_4 被 MOFs 材料固定 (Elevated temperature is conducive to the fixation of N_2O_4 by MOFs material)
- D. 转化为 HNO_3 反应的原子利用率约为 82% (The atom utilization rate of the reaction converted to HNO_3 is about 82%)

答案 (Answer): B

解析 (Analysis): 氮的固定是将游离态的氮转化为化合态的氮，该过程不属于氮的固定，A 错误；根据原理示意图知，“固定”过程体现超分子的分子识别功能，B 正确； $2NO_2(g) \rightleftharpoons N_2O_4(g)$ $\Delta H < 0$ ，温度升高，平衡向左移动，不利于 N_2O_4 被 MOFs 材料固定，C 错误；转化为 HNO_3 的反应是 $2N_2O_4 + O_2 + 2H_2O = 4HNO_3$ ，其原子利用率为 100%，D 错误。(Nitrogen fixation is the conversion of free nitrogen to combined nitrogen, and this process does not belong to nitrogen fixation, so A is incorrect. According to the schematic diagram of the principle, the “fixation” process reflects the molecular recognition function of supramolecules, so B is correct. $2NO_2(g) \rightleftharpoons N_2O_4(g)$ $\Delta H < 0$. When the temperature rises, the equilibrium shifts to the left, which is not conducive to the fixation of N_2O_4 by MOFs material, so C is incorrect. The reaction converted to HNO_3 is $2N_2O_4 + O_2 + 2H_2O = 4HNO_3$, and its atom utilization rate is 100%, so D is incorrect.)

18. 邻苯二甲酸二丁酯 (沸点为 $337^\circ C$) 可用作聚醋酸乙烯、醇酸树脂、硝酸纤维、乙基纤维素及氯丁橡胶、丁腈橡胶的增塑剂。利用 1-丁醇、邻苯二甲酸和浓硫酸共热反应，经过回流反应、蒸馏、萃取分液可制得邻苯二甲酸二丁酯粗产品，装置如图所示。(Dibutyl phthalate (boiling point $337^\circ C$) can be used as a plasticizer for polyvinyl acetate, alkyd resin, nitrocellulose, ethyl cellulose, neoprene and nitrile butadiene rubber. The crude product of dibutyl phthalate can be prepared by heating 1-butanol, phthalic acid and concentrated sulfuric acid, followed by reflux reaction, distillation, extraction and liquid separation. The device is shown in the figure.)



已知: ① 1-丁醇的沸点为 $118^\circ C$ ，受热易挥发，邻苯二甲酸的沸点为 $230^\circ C$ ；(Given: ① The boiling point of 1-butanol is $118^\circ C$, it is easy to volatilize when heated, and the boiling point of phthalic acid is $230^\circ C$;) ②



下列说法正确的是 () (Which of the following statements is correct? ())

- A. 装置 II 中 a 为进水口，b 为出水口 (In Device II, a is the water inlet and b is the water outlet)
- B. 装置 I 中回流反应的目的是减少物质的挥发，提高产率 (The purpose of the reflux reaction in Device I is to reduce the volatilization of substances and improve the yield)
- C. 回流反应后的混合溶液可先用饱和的氢氧化钠溶液洗涤除去混合溶液中剩余的酸，而后进行萃取分液 (The mixed solution after the reflux reaction can be washed with saturated sodium hydroxide solution first to remove the

remaining acid in the mixed solution, and then extracted and separated)

D. 用装置 III 萃取分液时, 将分层的液体依次从下口放出 (When extracting and separating liquid with Device III, the layered liquids are discharged from the lower port in turn)

答案 (Answer): B

解析 (Analysis): 蒸馏操作中, 冷凝管中的冷凝水应该下进上出, 故装置 II 中 b 为进水口, a 为出水口, A 错误; 由题干信息可知, 1-丁醇易挥发, 故装置 I 中回流反应的目的是减少物质的挥发, 提高产率, B 正确; 由于 NaOH 溶液碱性太强, 能使生成的邻苯二甲酸二丁酯完全水解, 故回流反应后的混合溶液可先用饱和的碳酸钠溶液洗涤除去混合溶液中剩余的酸, 且能溶解 1-丁醇和减小邻苯二甲酸二丁酯的溶解度便于分层, 而后进行萃取分液, C 错误; 分液时, 下层液体从下口放出, 而上层液体需从上口倒出, D 错误。 (In distillation operation, the cooling water in the condenser should enter at the lower port and exit at the upper port, so in Device II, b is the water inlet and a is the water outlet, so A is incorrect. According to the information in the question, 1-butanol is easy to volatilize, so the purpose of the reflux reaction in Device I is to reduce the volatilization of substances and improve the yield, so B is correct. Because NaOH solution is too alkaline, it can completely hydrolyze the generated dibutyl phthalate. Therefore, the mixed solution after the reflux reaction can be washed with saturated sodium carbonate solution first to remove the remaining acid in the mixed solution, dissolve 1-butanol and reduce the solubility of dibutyl phthalate for easy layering, and then extracted and separated, so C is incorrect. During liquid separation, the lower layer liquid is discharged from the lower port, and the upper layer liquid needs to be poured out from the upper port, so D is incorrect.)

